Katia Paola Victoria Escalera

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Summary

9th-semester Systems Engineering student (GPA 90.81) at TecNM – Instituto Tecnológico de Jiquilpan. Track: Intelligent Web-Mobile Application Development. Focused on frontend development (React, HTML/CSS/JS) and AI integration (TensorFlow/Keras) through FastAPI APIs. Seeking a professional internship in software development.

Education

Instituto Tecnológico de Jiquilpan (TecNM) — Jiquilpan, Michoacán
B.Eng. in Systems Engineering — *Aug 2022 – Present*
- Track: Intelligent Web-Mobile Application Development
- GPA: 90.81 / 100

Projects

ThermoNet | *Keras, FastAPI, Docker, HTML/CSS/JS* — Jan 2026 – Mar 2026
- Celsius → Fahrenheit converter built with a neural network in Keras (max error 0.0001 °F).
- REST API with FastAPI and a web interface with slider and real-time prediction. Dockerized.
- github.com/KPVE2004/thermonet

MNIST Pro | *Keras, FastAPI, Docker* — Apr 2026 – Jun 2026
- Handwritten digit recognizer with a CNN reaching 99.17% test accuracy.
- FastAPI backend and web interface to draw or upload digits, with a probability chart. Dockerized.
- github.com/KPVE2004/mnist-digit-recognizer

Driver Fatigue Detection | *MobileNetV2, FastAPI, OpenCV, Docker* — Jan 2026 – Jun 2026
- Computer vision system reaching 99.02% test accuracy.
- My role: Phase-2 fine-tuning, model evaluation, and frontend development with webcam access.
- Team project (3 members). github.com/KPVE2004/deteccion-fatiga-conductores

BC Boutique | *HTML, CSS, JavaScript, Figma* — Jan 2025 – Jun 2025
- Web page for a fashion boutique, designed from scratch in Figma (18 screens).
- Built the full frontend: catalog, product detail, cart, favorites, login, search, and mobile menu.
- github.com/KPVE2004/bc-boutique

KIA Compiler | *HTML, CSS, JavaScript* — Jan 2025 – Dec 2025
- Web-based didactic compiler with lexical, LR syntactic, and semantic analysis.
- Designed the full UI (editor, result tabs, toolbar) and the visual identity (logos).
- Team project (3 members). github.com/KPVE2004/compilador-kia

Minimal Conversational Agent | *LangChain, Groq, FastAPI, Django* — Aug 2026 – Present
- Chat agent built with LangChain and Groq. Redesigned the chat UI and added a "typing" indicator. github.com/Omarvl-XCoder/Agente-Conversasional-Minimo

Fleet Management with AI | *Django, FastAPI, PostgreSQL, MongoDB, Redis Streams* — Aug 2026 – Present
- Microservices-based fleet monitoring system with AI-based fatigue detection. Development Team member in a Scrum team (JIRA). github.com/AngelCanelaaa/gestion-flotillas-ia

Technical Skills

Frontend: HTML5, CSS3, JavaScript, TypeScript, React, Tailwind, Bootstrap
Design & Mobile: Figma, Canva, React Native, Kotlin
Backend & AI: Python, Java, FastAPI, Django, TensorFlow, Keras, OpenCV
Databases: SQL, PostgreSQL, MongoDB, SQLite, Firebase, SQL Server, phpMyAdmin
Tools: Git, GitHub, Docker, Linux

Languages

Spanish: Native
English: B1 (CEFR)

Certifications

Foreign Language Completion Certificate (English B1) — Oct 2024
- Instituto Tecnológico de Jiquilpan (TecNM)

Recognition for Completing English Modules (450 hours) — Dec 2024
- Tecnológico Nacional de México / ITJ